

MICHAEL K. MIHAILOV

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HIGHLIGHTS OF QUALIFICATIONS

Software Development: Full-stack web, desktop, game, and agent-based simulation development

Algorithms & Problem Solving: Algorithm design, computational modeling, and complex problem solving

Research: Undergraduate research in agent-based disease modeling and simulation

Performance Optimization: Achieved a **160× simulation speedup** and **45% codebase reduction** through optimization

Collaboration & Leadership: 4-person development teams using GitHub and Agile/Scrum

Technical Communication: Presented research findings at the Lewis University SURE Symposium

SUMMARY OF TECHNICAL QUALIFICATIONS

LANGUAGES: (By Proficiency)

- Java, Python, NetLogo, C#, JavaScript, TypeScript, C++

FRAMEWORKS & LIBRARIES:

- React, Express.js, FastAPI

TECHNOLOGIES:

- HTML, CSS, Node.js, WebSockets, Unity, MySQL, SQLite

TOOLS & PRACTICES:

- GitHub, Agile/Scrum

CORE CONCEPTS:

- Object-Oriented Programming, Data Structures & Algorithms, Multithreading

EDUCATION

Bachelor of Science in Computer Science
Lewis University, Romeoville, IL

Expected: Dec 2027
GPA: 4.0/4.0

PROJECTS

Classroom Scheduling System — Software Engineer

Dec. 2023 - May 2024

JavaScript, React, Node.js, MySQL, Material UI, WebSockets, Raspberry Pi | [GitHub](#)

- Developed full-stack scheduling features as part of a 4-person software engineering team, enabling teachers to upload and manage classroom schedules through an administrative web portal
- Implemented backend logic to process teacher-submitted schedules and deliver schedule data to classroom display clients
- Built real-time WebSocket communication between clients and the server, allowing schedule changes to propagate immediately without page refreshes
- Developed Raspberry Pi-powered classroom displays showing live period timers, class information, teacher notes, and optional weather data
- Deployed application across multiple classrooms at Naperville Central High School; system continues to be used and maintained

Poster Generator — Sole Developer

Apr. 2023 - May 2023

Java, Java AWT, Recursion, ArrayLists, Image Processing | [GitHub](#)

- Developed a Java application for programmatically generating high-resolution posters from PNG images
- Designed and implemented a recursive Apollonian gasket algorithm from scratch to generate progressively smaller circles within packed geometric regions
- Implemented image-processing operations including arbitrary-angle rotation, mirroring, color negation, edge detection, grayscale conversion, and image resizing
- Integrated image transformations into the generated geometric layout, placing independently manipulated images within the gasket's circles

The Adventures of Pecan and Suede — Lead Programmer

Mar. 2022 - May 2022

C#, Unity, WebGL, GitHub | [Demo](#)

- Led programming for a 4-person team developing a cooperative 2D puzzle-platformer, implementing the majority of the game's technical systems
- Developed core gameplay systems including player movement, game-state management, collision handling, interactive UI, and the game loop using Unity and C#
- Programmed approximately 20 progressively challenging levels designed for two players sharing a keyboard
- Integrated gameplay, level progression, and user-interface systems into a browser-deployable WebGL build

RESEARCH

Undergraduate Researcher

Summer 2026

Summer Undergraduate Research Experience (SURE), Lewis University, Romeoville, IL

- Refactored and streamlined a NetLogo agent-based model simulating *C. difficile* transmission across hospital and long-term care facilities, reducing codebase size by 45% while adding new functionality
- Optimized simulation performance by identifying and eliminating computational bottlenecks, reducing runtime for one year of simulated time from 10,030 seconds to 62 seconds (160x speedup)
- Developed a Python-based automated calibration system using CMA-ES and parallel NetLogo instances to efficiently tune model parameters against target outcomes
- Implemented and evaluated disease-control interventions including admission screening, room isolation/quarantine, and vaccination, analyzing their effects on simulated infection incidence
- Identified and corrected numerous logical and behavioral bugs in the existing simulation, including healthcare-worker scheduling, resident activity, bathroom interactions, and simulation stability
- Presented research findings and experimental results at the Lewis University SURE Symposium and continued development of the model through MATH 47500 (Mathematics Research)

HONORS AND AWARDS

- Lewis University Presidential Scholarship (2025-Present)
- Illinois State Scholar (2025)
- Senior Student of the Year in Computer Science, Naperville Central HS (2025)